



Research on the Usability Design of HUD Interactive Interface

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Abstract. In the context of the Industry 4.0 era, with the rise of artificial intelligence technology and autonomous driving of automobiles, automobile companies are carrying out extensive technology research and development, product deployment and commercial layout around technologies such as autonomous driving, car networking, and human-computer interaction. In the development stage of automobile intelligence, a large amount of information and data such as entertainment and social interaction enter the in-vehicle system. The interactive information inside and outside the vehicle collected with physical components such as on-board sensors enriches the driver's driving experience to a certain extent, but at the same time it is easy to let The driver is distracted and affects driving safety. Head-up display (HUD), as a new driving assistance safety system, is considered to be the main way of presenting in-vehicle information in the future. However, the HUD information display system supported by technology lacks specific research on the design of human-computer interaction interface. This article will clarify the specific design based on the eye movement experiment to achieve the purpose of usability.

Keywords: HUD · Safe driving · Interface interaction

1 Research Background

In the context of the Industry 4.0 era, with the rise of artificial intelligence technology and autonomous driving of automobiles, automobile companies are carrying out extensive technology research and development, product deployment and commercial layout around technologies such as autonomous driving, car networking, and human-computer interaction. In the development stage of automobile intelligence, a large number of information and data such as entertainment and social interaction enter the in-vehicle system. The interactive information inside and outside the vehicle collected with physical components such as on-board sensors enriches the driver's driving experience to a certain extent, but it is easy to make the person be distracted and affect driving safety.

As a new type of driving assistance safety system, the head-up display is considered to be the main way of presenting in-vehicle information in the future. However, the HUD information display system supported by technology lacks specific research on the design of human-computer interaction interface. This article will be based on eye movement experiments with the help of the human-machine driving information transmission relationship model and other methods are clearly designed to achieve the purpose of usability.

2 Interface Visual Design and Application

2.1 Interactive Interface Design Principles

Compared with traditional interaction methods, the interaction design of automobiles has the biggest difference that it changes based on different situations. The car is always in a changeable and complex situation, far different from the stable interactive environment in the past. Therefore, we need to combine interface design principles with specific driving environment characteristics, and use this as a basis to design automobile interactive interface principles.

- (1) **Rationalization Principle of Information Selection Layout.** According to the research results of vehicle ergonomics, about 60% of the visual attention during driving will be focused on maintaining the state of the vehicle and coping with emergencies.

Therefore, the vehicle interface should not occupy too much visual channels, but needs to ensure that users can quickly Get information and perform operations. therefore, in the limited visual channel, it is necessary to classify the information, divide the information priority, retain the necessary information according to the frequency and importance of use, and discard irrelevant information barriers. At the same time, in the information layout, more use of graphics and text forms, as far as possible to enable users to complete most of the information acquisition and operation behavior on the main page.

- (2) **Information Feedback Principle.** To change the default, adjust the template as follows. The information feedback principle is based on the feedback mechanism to guide the interaction design, in order to ensure that users get instant feedback in the context of use, confirm operations, and improve the success rate of operations. Context-based vehicle interaction interface information is complex and variable, so it is particularly important to feedback this information immediately.

The specific performance of information feedback is: timely feedback the changes and status of objects and programs; provide feedback to remind users whether there are errors, and help users correct in time during operation; special symbols and graphics should be used to indicate where there is a need to increase attention The meaning of it; when the user is resting or working, the message should be used to remind the user in a softer way.

- (3) **The Principle of Interface Simplicity.** In today's choice of car interaction methods, the functions of physical buttons and knobs are gradually shifted to touch-sensitive interactive screens, which undoubtedly increases the difficulty of interface design. A simple interactive interface with clear instructions can shorten the human-computer interaction time and increase the accuracy of the interactive content, ensuring driving safety.

The interface design should be concise but not simple. The semantic design of information should clearly express icons, scales and numbers to avoid ambiguity in visual understanding. The design and arrangement of buttons should consider the consistency of coding, which is conducive to memory and enables users to perform blind operations after familiarization.

2.2 Interactive Interface Content

Refer to the relevant content of the vehicle driving system, the system information can usually be divided into five categories: basic vehicle information, navigation system, driving safety warning, social entertainment information, and auxiliary imaging system, as shown in Table 1 shows:

Table 1. Assisted driving system information classification

Basic vehicle information	Navigation system	Driving safety warning	Social entertainment information	Auxiliary imaging system
Speed	Road indication trajectory	In-car safety warning information	Bluetooth button control	Reversing camera system
Rotating speed	Distance to destination	Obstacle information	Voice control	Visual blind zone imaging
Mileage	Road name	Lane information	Gesture Recognition	
Water temperature		Identification information	We Chat	
Oil temperature			SMS	
			Music	

However, in addition to the basic principles of interaction design, the interface design should also focus on the cognitive load of the driver during the driving process and the real-time dynamics and multi-dimensional complexity of the driving situation in the selection of content. Excessive occupation of visual channels and receiving cumbersome interactive information will often lead to a certain difference between the conceptual model of the system and the user’s mental model, which will affect driving safety.

Therefore, it is necessary to construct a hierarchical HUD security prompt information model. This study is based on the principle of information classification, and the information priority is divided according to the importance of interactive content and frequency of use, as shown in Fig. 1 below. The content with higher priority is status information display and context information display. The status information display interface includes basic driving information, warning information, and social entertainment information; the context information display interface includes collision avoidance

instructions and navigation instructions. Therefore, the interface usability design will be expanded on such interactive interface content.



Fig. 1. Information visualization chart

2.3 Interface Usability Design

The interface information of the assisted driving system is closely related to the driving environment and user driving behavior. In the design process, it is necessary to comprehensively consider the font, functional layout and color tendency of the visual elements.

- (1) **Font.** MIT’s Age Lab, the University of New England Transportation Center, and Monotype conducted a study on the reading efficiency of sans-serif fonts due to different line spacing widths and different font thicknesses. It was found that Square Grotesque fonts increase the reading time by 12% compared with Humanist fonts, which makes cars in the braking distance when driving at the same average speed differs by 50 feet, and a distance of 50 feet means the difference between safe driving and a serious accident. Monotype proposes to use human-oriented sans-serif fonts, among which Roboto fonts are generally considered to be the most accurate and efficient fonts for reading, and have been verified by Google to read information in the car, as shown in Fig. 2.



Fig. 2. Information priority classification

- (2) **Information Layout.** The functional layout of the HUD interface is based on the content of the interface information. After priority classification, the five functional module information of basic driving information, anti-collision instruction information, navigation instruction information, warning information and social entertainment information are obtained. In order to further determine the location of the module, the study uses the driver to look straight ahead as the first perspective, and divide the interface into regions, as shown in Fig. 3:

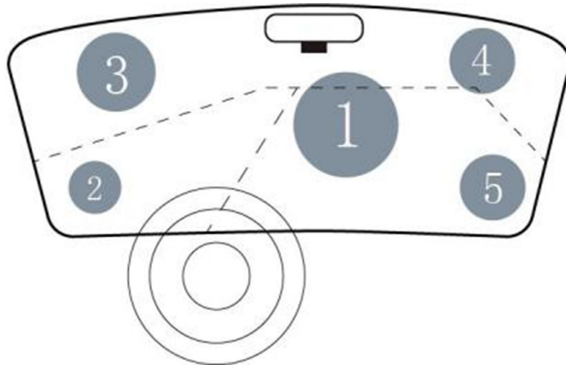


Fig. 3. Module area division

First of all, the middle position of the interface occupies most of the driving angle of view. This area (position 1) mainly presents basic road conditions, and is associated with the anti-collision indication information, such as detecting the distance, speed, safety distance, pedestrians and other information of the vehicle ahead.

Secondly, the information that needs to be recorded is the basic driving information. In the simplest mode, this area mainly presents the vehicle speed value and unit information. It has a small area and does not need to occupy too much area, so it is more appropriate to place it in position 2.

Furthermore, the area with a larger interface area and more scattered information is the navigation indicator information module, which has strong display flexibility, and most systems support user-defined display content; the similar area is the warning information module, it contains random auxiliary driving information such as road speed limit information, cruise control information, vehicle distance prompts and other information. The purpose of both is to display the information in the driving process more intuitively and concisely, avoiding the occurrence of dangerous situations due to distraction. Therefore, these two module areas are suitable for 3 and 4 parts.

Finally, place the social entertainment information area with the lowest priority at position 5, which is in the lower right quadrant in human visual cognitive habits, and is in the area with the lowest recognition rate. This requirement is just in line with the low frequency and unimportant information characteristics of social entertainment information in driving.

- (3) **Color Tendency.** Color is very important in interface design. In completing visual tasks, color coding is beneficial to enhance the efficiency of information recognition, thus shortening the information processing time. The visual interface of the HUD assisted driving system is a virtual image projected on the front windshield of the vehicle. The visibility of interface elements will be affected by factors such as light, weather, road conditions, and environment. Therefore, it is necessary to fully ensure that the clarity of the element display is not disturbed when selecting colors.

At the same time, this research needs to select the color of HUD in combination with the traffic safety color system. In the road traffic safety color system, most of the high-visibility color designs and applications are based on green, yellow and red. Among them, the red system means prohibition and stop, the green system means safe and feasible; the yellow system means warning and buffering to distinguish different safety prompts.

Refer to the color matching of the existing HUD in the current market, as shown in Fig. 4. The HUD interface design of this study uses blue as the main color, and red, yellow, and green as auxiliary colors to display information prompts. The specific basis will be explained in the eye movement experiment verification in Sect. 3.

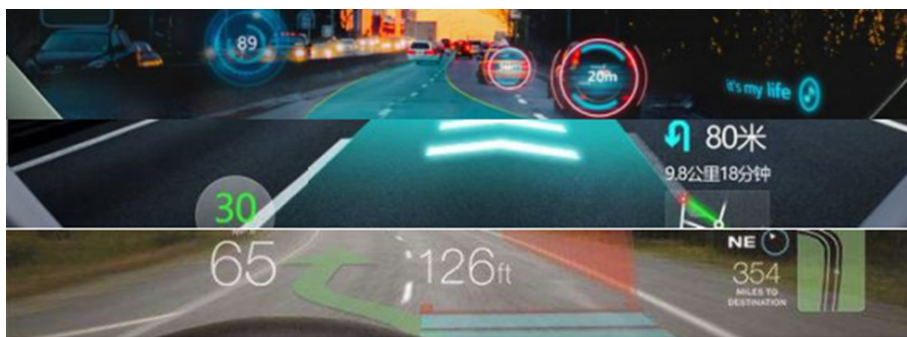


Fig. 4. HUD Color Matching

3 Eye Movement Experiment and Verification

3.1 Experimental Study

Purpose. The core task of the HUD assisted driving system is to assist the driver to effectively complete the driving task and improve driving safety. The study focuses on the specific design of the usability of the interactive interface, and clarifies the size, color and location of the interface elements. According to the design description in the third part, in order to verify the effectiveness of the HUD system design, the purpose of this experiment has two points:

Experiment 1: Through the accuracy test of the interface safety prompt information, it is verified that the HUD system visual interface of the information layout described in the third part has the best cognitive performance.

Experiment 2: Through the sensitivity test to potential hazards in complex driving situations, it is verified that the color selection of the HUD system visual interface described in the third part has the best response ability to the driver's danger.

(1) Experimental Method. This experiment conducts an objective evaluation of driving safety from the accuracy of interface cognition and the driver's ability to react dangerously. It mainly adopts eye movement experiment method and spoken language analysis method.

Eye Movement Experiment: Aiming at the interface information cognitive efficiency and driver's risk response ability test, the eye movement experiment method is mainly used to record the information cognitive situation and driving behavior response of the subjects under different information layout and color selection interfaces.

Oral Analysis: For the test of the accuracy of interface information cognition, in addition to the eye movement experiment, the oral analysis method is mainly used, which can externalize the results of the implicit process such as the user's information cognition stage and cognitive strategies. After the experiment process is over, the subjects are asked to give oral answers to the specific information content that appears, so as to judge the driver's information cognition efficiency under different information layouts.

(3) Experimental Variables

Independent Variable:

Experiment 1: Information layout of system interface

Experiment 2: HUD system visual interface color

Dependent Variable:

Experiment 1: Cognitive performance

Experiment 2: Risk factor sensitivity

3.2 Experimental Design

- (1) **Participant Selection.** In order to prevent the participants from having insufficient knowledge of the driving interface information, using experience, and lack of basic information cognitive ability of the HUD system functional interface, this experiment selects participants with a certain driving experience, and minimizes the participants' exposure to the HUD. Factors such as cognitive bias caused by system unfamiliarity. There are 12 participants, 9 males and 3 females, with driving experience in the range of 2–10 years.
- (2) **Experimental Material.** The experimental instrument of the eye movement experiment adopts the portable eye tracker of Tobii Pro's X3–120 model, which can record the trajectory of human eye movement, fixation point, fixation time and other characteristics. At the same time, according to different experimental purposes, this study also selected and produced two sets of experimental materials according to experimental requirements, as follows:

Experiment 1: Make HUD visual interface materials with different information layouts. According to the difference of independent variables, the experiment produced three kinds of experimental materials with different information layouts, as shown in Fig. 5 below. The different information layout interfaces displayed by the experiment materials were sequentially changed, and different subjects were required to memorize the specific information content and answer related questions after the experiment. The questions are as follows:

- In the current situation, how far is the vehicle from the position of the vehicle ahead?
- In the current situation, what is the driving speed of the vehicle?
- In the current situation, how far is the left turn from the navigation prompt?
- In the current situation, what kind of warning message is there for the distance ahead?
- In the current situation, who is making the voice call?

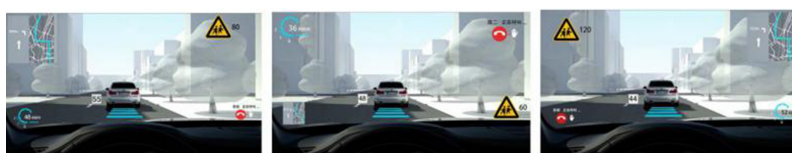


Fig. 5. Information layout interface material

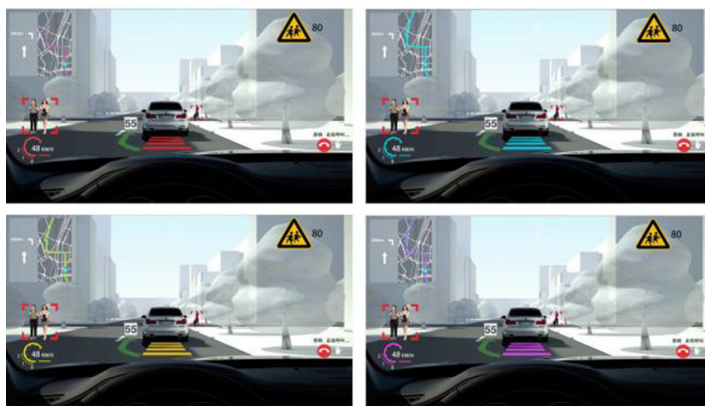


Fig. 6. Visual color interface material

Experiment 2: Make different HUD visual color interface materials. According to the difference of independent variables, the experiment produced four different color interface materials, as shown in Fig. 6. The experiment requires the subjects to freely watch the assisted driving interface of different colors in turn, and arrange relevant risk factors

in the actual situation in the interface, such as: Side pedestrians, distance to the vehicle ahead, lane location, etc. After the experiment, the gaze hot zone of the eye movement experiment parameters was observed to record the subjects' sensitivity to risk factors during the test.

(3) Cognitive Performance Evaluation

Cognitive Performance Evaluation: The inspection and evaluation of the cognitive performance of visual interface information mainly focus on the speed and accuracy of cognitive information. In this experiment, the latter is selected as the cognitive performance indicator, and the correct rate of the driver's question answer after the completion of the test is defined as the cognitive accuracy rate. Failure to answer or failed to answer is defined as an error. Finally, the number of correct cognition and the ratio of the total times is the cognitive accuracy rate.

Risk Sensitivity Evaluation: Improving driving safety is the purpose of the HUD assisted driving system. Therefore, prompting of potential hazards in complex driving situations is an important part of HUD. For the driver, the degree of subjective sensitivity to hazard warnings determines the ability to react to hazards. Therefore, this experiment defines the driver's hazard sensitivity as an important indicator to verify driving safety efficiency under different visual color interfaces.

3.3 Data Processing and Analysis

- (1) **Cognitive Performance Analysis.** Through the experiment, a pair of 12 subjects' cognitive efficiency was recorded and analyzed, and the result of the accuracy

Table 2. Information layout interface and cognitive accuracy

Participant serial number	Material number	Number of correct answers (times)	Cognitive accuracy (%)
1	Material 1	5	100%
2	Material 2	1	20%
3	Material 3	3	60%
4	Material 1	3	60%
5	Material 2	2	40%
6	Material 3	4	80%
7	Material 1	5	100%
8	Material 2	1	20%
9	Material 3	3	60%
10	Material 1	4	80%
11	Material 2	1	20%
12	Material 3	3	60%

of the information layout on the cognition was calculated, as shown in Table 2. From the data in the figure, the interactive interface of different information layouts. There is an impact on the efficiency of knowledge. The cognitive accuracy rate (85%) of the information layout interface reflected by the material is better than the other two groups (25% and 65%), indicating that the information layout HUD visual interface designed in this experiment is conducive to the same situation, the reading and understanding of driver information at the same time conforms to the law of driver information recognition.

- (2) **Risk Sensitivity Analysis.** Through the experimental records of the 12 subjects in Experiment 2, the driver's eye movement hot zone diagrams under different visual color interfaces are obtained, as shown in Fig. 7. Under no experimental task, with the same experimental situation and equal experimental time, driving the gaze of the interactive interface of different visual colors is different.

According to the evaluation index, the visual interface with blue as the main color tends to pay the most attention to potential hazards in the road in the driving task, especially for pedestrian elements, showing red hot spots; the interface with yellow as the main color has the gaze of the vehicle in front. The highest, low attention to pedestrians; the red and purple color interface hot zone is messy, indicating that the driver's gaze is scattered during driving and driving danger is prone to occur.

From the analysis results, it can be seen that under the premise of the same driving situation, the HUD system with a blue-based visual interface can improve the driving safety of the driver, has a good role in assisting driving, and is more sensitive to danger in driving other HUD vision systems.

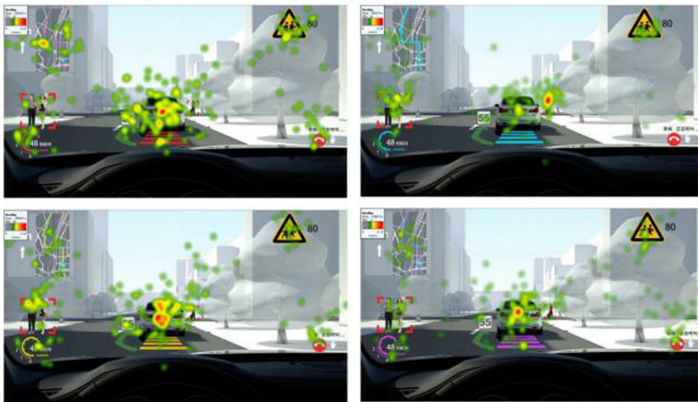


Fig. 7. HUD color matching

3.4 Experimental Results

Through objective experimental evaluation, the hypothesis of the purpose of this experiment was fully verified:

1. In the performance evaluation of information cognition in experiment 1, the cognition accuracy rate of the information indicators of each group of the experiment was analyzed. The data results showed that the information cognition accuracy rate of material 1 was better, which verified the rationality of the information layout design results. That is, the HUD visual interface with this information layout has the highest cognitive performance.
2. In the evaluation of the risk sensitivity of experiment 2, the color interface of each group of the experiment was analyzed for the risk sensitivity. The data results show that the HUD system with a blue-based visual interface can play a good auxiliary role and improve the driving safety of drivers. This color-oriented HUD visual interface has the highest color sensitivity.

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